

## Project Description

### *Attracting and Retaining Undergraduates to Engineer the Built Environment*

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#### **I. Proposal Overview**

This proposal from Columbia University's Departments of Civil Engineering & Engineering Mechanics and Earth & Environmental Engineering in collaboration with the Columbia Center for New Media Teaching & Learning is designed to address the fundamental problem of attracting and retaining undergraduates to civil & environmental engineering. This problem plagues many traditional engineering programs throughout the United States, so we expect this project and its results to have notable implications for the reform of engineering education.

We will develop a working prototype that simulates urban scenarios, named **OPEN PLATFORM FOR TEACHING INTEGRATED MODELING AND URBAN SIMULATION (OPTIMUS)**; the OPTIMUS prototype will integrate a three-dimensional interface of urban environments with underlying databases and models that represent components or systems within these environments. The prototype will become the central feature of a re-designed undergraduate curriculum that exposes students to the issues and variables of the built environment early in their undergraduate experience. By presenting realistic engineering scenarios to students as either freshmen or sophomores, we expect this exposure to “demystify” the discipline, so students have a concrete understanding of the roles engineers play in the built environment. More importantly, students receive an introduction to problem solving that emphasizes an integrated and systems approach, so they recognize that the application of engineering expertise requires synthesis of topical knowledge.

Our working hypothesis is that this early exposure to the context of civil & environmental engineering problems will: (a) motivate students to enter and complete our programs of instruction, (b) deepen student appreciation of the fundamental theories of mathematics and mechanics since they understand the significance of these theories to their ultimate ability to analyze real-world problems, and (c) better prepare students to analyze complex, multi-disciplinary problems.

The remaining sections outline the circumstances that motivate this project, our project team, specific objectives, and approach.

#### **II. Introduction**

Civil Engineering is arguably the second oldest profession. Indeed, formal engineering education traces its roots to this profession. A rich history of developing and managing global physical infrastructure (buildings, bridges, transportation systems, dams, treatment plants, waterways) underscores the ever changing and growing domain of the field. Disciplines such as environmental engineering have emerged from this tradition, as the societal emphasis on sustainability and environmental protection has emerged as an objective. This field differs from other engineering disciplines in a) breadth, b) spatial and temporal scales of operation, c) the emphasis on products for society rather than individuals, d) the predominance of the public sector in the field, e) the direct and lasting impact and interaction of its products with landscape and social structures, and f) the role the profession and its products play in enabling economic growth and the successful functioning of other fields. As society recognizes that planetary and regional change is inevitable, and our responsibility lies in successfully managing such developments, the important role of the Civil and Environmental Engineer (CEE) in the urban, regional and national setting becomes clear. Unfortunately, undergraduate degrees in civil engineering have declined by nearly 15% nationwide over the past five years while degrees granted in environmental engineering have held relatively constant (ASCE, 2001). Underrepresented populations such as women and African-Americans have fared marginally better, but the percentage of degrees granted to these two groups remains quite low.

Technology clearly has a profound impact on society and on education. For example, classes on drafting and surveying, staples of civil engineering education thirty years ago, have largely vanished. Biological rather than physical processes now dominate waste treatment. The list of topics a CEE “must know” has both changed and increased dramatically decade by decade. Wulf (1998), while arguing for urgent reform in engineering education notes that the “half-life” of

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engineering knowledge is estimated to be in the range of 2.5 to 7.5 years. He questions the value of a narrow specialization when *“engineered systems are becoming larger and more complex, and involve components and processes from many more fields of engineering”*. Likewise, Bordogna (1998) stresses the importance of systems education for CEEs instead of reliance on an ever increasing and changing collection of courses on “reductionist engineering science”. Roesset and Yao (2000) note that not only has the basket of skills necessary to be a CEE changed, changes in technology have fundamentally changed the engineering method, and the type of undergraduate education needed. Traditionally, CEEs were trained to be *“efficient solvers of routine problems”*, using a mix of empirical rules and first principles. Today, software packages allow the parametric investigation of a range of design alternatives in minutes supplanting years of traditional office experience. Changes in the undergraduate CEEing curriculum, particularly in the first two years, have not been particularly radical or responsive to this changing situation. An apparent reduction (Walesh, 2000) of about 20 semester credits in the undergraduate requirement over the last 50 years appears to have come largely at the expense of field and laboratory experience, and in courses oriented towards practice. Courses on Civil Engineering Systems were introduced and popular in the 1970s, and yet are rare today - an apparent victim of their evolution as skill courses in optimization theory and its application. These trends are remarkable given the attendant technological and information explosion over this time period, and the emerging importance of using the increasing amounts of data to understand, resolve and manage spatial interactions between physical infrastructure and the environment.

While technological innovations have focused on novel learning modules for K-12 programs, and to some extent for graduate science education, the basic instructional model in undergraduate engineering education has only recently received attention. Roesset and Yao (2000) observe, *“most teaching at present is abstract (principles rather than applications), verbal, deductive (going from general axioms to*

*applications), and sequential.”* The lecture format dominates. Coupled with a lack of exposure to problem formulation and solution in the first two years, and the lack of attention to the *“people serving dimension”* of the profession, many students leave programs during this period (Walesh, 2000; Grigg, 2000). Remarks documented by Walesh from student exit interviews include *“I want to be involved in something creative”*; *“Engineering has nothing to do with people”*; and *“I want to make a difference”*. Research on classroom learning (see Campbell, Burge and Kirkup) suggests that the traditional, lecture/abstract approach is more successful with male students, and that women prefer *“group discussions, simulations, panels and other activity based learning”* (Campbell, <http://www.atl.ualberta.ca/articles/idesign/learnhar.cfm>). An emphasis on communication skills and team functioning for engineers also emerges uniformly in most recent opinions as to engineering education needs. Roesset and Yao (2000) report on a number of recent University initiatives that seek to address this situation. Some common elements across the programs reviewed by them (Louisiana Tech, Texas A&M, Maryland, Air Force Academy and Colorado) are (a) students work as teams on design problems often with practitioner participation from the very early stages; (b) participatory activities in the classroom replace part of the lecture period; and (c) conscious integration of material across classes. The Foundation Coalition headed by Texas A&M reports several encouraging results (<http://coalition.tamu.edu/Zope/success>) from the implementation of these *Active and Cooperative Learning* (ACL) methods (<http://coalition.tamu.edu/Zope/acl>), including much higher retention rates of both male and female students, a more positive student attitude, and higher scores on the fundamentals of engineering examination. Despite these welcome trends, Wulf (1998) notes that the opportunity for Information Technology in shaping the undergraduate engineering classroom *“through its ability to provide access to vast information sources, to support discovery based educational experiences safely, and to more aggressively support peer-to-peer education”* is largely untapped. The use of virtual reality and

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simulation could make engineering methodology accessible in a holistic setting for education in much the same way that higher level programming languages, and graphical, numerical and statistical libraries make developing software considerably easier today than 30 years ago. Indeed, they provide a vehicle for imparting technological or engineering literacy to non-majors, and a potential recruitment vehicle in the early years.

The preceding discussion indicates the importance and challenge of reforming undergraduate CEEing education. A focus on active and cooperative learning methods facilitated by simulation and visualization technology, and on an integrated four year curriculum is indicated. Some specific areas of interest are now outlined. Nowhere are CEEing problems more pronounced than within our nation's urban centers that host an array of contemporary challenges. Congested transportation corridors impact air quality; aged water distribution pipes leak and deplete supply reservoirs; combined sewers overflow during heavy rainfall and release untreated sewage into natural waterways; development activities contribute to non-point source water pollution; and inefficient buildings seep conditioned air and increase energy consumption; ecological impacts lead to calls for removing dams that supply water and energy and provide flood control for population centers; roadways and bridges continuously deteriorate due to environmental and traffic exposure; inadequate structures await destructive earthquake and hurricane forces; emergency response and medical facilities are often crippled at times of disaster by non-structural building component failures (i.e. elevators, power, etc.).

These problems are the result of the interaction of numerous factors that impact our urban environments. Population change and infrastructure deterioration are two of the more salient variables since they impact the demand for natural resources and the serviceability and reliability of infrastructure systems. Numerous studies and publications (*America in Ruins*, *Fragile Foundations*, *ASCE's Report Card for America's Infrastructure*) have cataloged the deterioration of our national infrastructure, and

regardless of the merit given to various estimates of infrastructure needs, a prolonged effort to renew and replace components and sub-systems of our national infrastructure is a logical expectation. Further, this cycle is perpetual; today's challenges are replicated by similar challenges in the future. Essentially, the century old infrastructure systems in many of the nation's urban centers are approaching the end of their first lifecycle. The future includes second, third and fourth cycles. The duration of these projects and their spatial arrangement can significantly impact social structure, which in turn can impact the effectiveness of the structure. Further, changes in climate, and in societal values add elements of uncertainty that need to be considered in the renewal of infrastructure. In the past, infrastructure development and environmental management were centralized functions of single manager groups, and economic factors (e.g., benefit-cost analysis) dominated the design process. Today, collaborative decision making with multiple stakeholders and geo-spatial considerations are emerging as the norm. Many of these factors are best introduced early in the curriculum since they provide the context for the technical knowledge and do not require sequential skill material.

In short, the *thrust of this proposal is to take a fresh look at the undergraduate education of the CEE* and to develop technological innovations that support curricular redesign and student learning. The proposed technological innovations are designed to: a) make the analysis of infrastructure and environment in the proper social and spatial context accessible, b) emphasize the use and collection of data, c) stimulate in-depth and creative research, d) involve students in solutions to community problems so that they appreciate their historical and cultural context, and e) generate the student excitement that this field warrants. Since CEEs are predominantly involved in the public sector, and design and manage for society, relevant experience for those functions needs to be integrated in the academic curriculum. The first two years of the program are considered especially important to define a cohesive curriculum to replace what is often perceived as a loose collection of requirements for graduation.

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#### III. Background Information

**Institutional Background.** Columbia University has separate departments of Civil Engineering and Engineering Mechanics (CEEM) and Earth and Environmental Engineering (EEE) that work closely together. A recent University wide strategic initiative on “*Earth and Environment*” formalized through the Columbia Earth Institute (<http://www.earthinstitute.columbia.edu/>) provides an umbrella for the coordination of a number of academic and research programs. The role of CEEM and EEE is to facilitate the analysis and development of technical solutions to earth and environmental problems, in concert with the School of International and Public Affairs (SIPA), the Department of Earth and Environmental Science (DEES), the Department of Ecology, Evolution and Environmental Biology (E3B) and Barnard College.

While the CEEM and EEE programs at Columbia have been quite traditional, undergraduate students get comprehensive exposure to the basic science and liberal arts curriculum through a common core program for all majors for the first two years. Consequently, the primary contact that undergraduates have had with their engineering department in the first two years has been quite limited. Recently, both CEEM and EEE have introduced introductory level survey courses in the freshman year (*Design of Buildings Bridges and Spacecraft*, and *Earth Resources and the Environment*) that have been very popular relative to the size of each department. Despite the relative popularity of the courses the ‘yield’, measured in terms of the proportion of the class who remain as majors is low.

**Recent Initiatives.** The two departments secured an internal grant last year to initiate a comprehensive reformulation of the undergraduate curriculum to better address integrative, complex, earth systems problems. New faculty (including Garvin and Lall) with relevant experience were hired during this period, and the initiative is strongly supported through the departments and upper administration. This internal grant accounts for

cost-sharing applied towards achieving the goals of the proposed project.

One of the objectives of curriculum revision is to provide technical literacy to non-majors, and to recruit motivated students already admitted through a competitive process into our programs. Another goal is to position Columbia as a leading provider of state of the art instruction using multiple media and delivery mechanisms on issues pertaining to infrastructure and environmental management – areas identified as the primary societal challenge for the 21<sup>st</sup> century and designated as focal points of the University’s mission to its community. In interviews with entering freshmen, we have learned that this direction and the mandate of the engineering programs to develop solutions to such problems is of particular interest to women students. This interest is also manifest in the gender distribution of enrollment in the existing related classes. All students expressed a strong interest in early exposure to and involvement in the analysis of local, regional and global problems, and in working on projects with practitioners. Work on developing a new curriculum in this direction is underway; key elements are discussed later in the proposal.

**Engineering & CCNMTL Partnership.** CEEM & EEE and the Columbia Center for New Media Teaching and Learning (CCNMTL) have formed a partnership to produce a working prototype called OPTIMUS as an integral component in the redesign of the undergraduate curriculum, and to broadly disseminate resulting tools and outcomes. The Departments and CCNMTL bring to this project a combined expertise and experience that ensures its successful completion, dissemination, and evaluation. The CEEM and EEE Departments, with their staff of scholar-teachers and researchers, provide rigorous scholarship and engineering insight, while CCNMTL, with its 60 programmers, designers, education technologists, and research and development staff, has expertise in knowledge architecture, interface and pedagogical design, and web-based delivery systems that will support the creation of the simulation and its pedagogical use.

CCNMTL, founded in 1998, was created to better facilitate just the type of educational ideas

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brought forward in this proposal. Its mandate is to forge partnerships with faculty to help them develop promising technical solutions in order to realize educational goals. In the few years since its inception, CCNMTL has pioneered the development of various web-based simulations intended to serve both the classroom and the online learner. Notable relevant and exciting educational simulation tools developed by CCNMTL include:

- *Brownfield Action* simulator which provides a learning environment for developing skills in environmental science.  
(<http://www.ccnmtl.columbia.edu/services/showcase/bfa.html>),
- *Heart Simulator*, developed in collaboration Columbia's College of Physicians and Surgeon  
(<http://www.ccnmtl.columbia.edu/projects/listing/heart/>),
- *NewsSim*, developed with Columbia's School of Journalism for their required "Reporting and Writing" class  
(<http://www.ccnmtl.columbia.edu/projects/newssim/>).

**Educating with Simulations.** The utility of an open visualization, simulation and gaming environment that can allow students to explore infrastructure development and environmental scenarios, became apparent during our early discussions on how to represent social factors, natural hazards, degradation and the interaction of infrastructure, landscape, ecology, climate, social and institutional factors. Simulation and role-playing games such as *SimCity* and *SimEarth* appeal to students since they can quickly appreciate context and get feedback as to policy decisions or spatial structures they introduce (see Box 1). We have used *SimEarth* to introduce coupled modeling of planetary systems in an introductory Earth Science class. The game was complemented with traditional learning material on planetary energy balance, atmospheric composition, etc. Students explored issues related to anthropogenic climate change by altering CO<sub>2</sub> sources and sinks, and by changing attributes of the biological system. Concepts of residence times in different

#### **Box 1 – Existing SimCity Images and Scenarios**

*SimCity* has realistic visual images of actual urban settings already developed. For instance, the graphic below shows downtown Sacramento, and the geographic and urban character of the city is evident. In addition, natural hazards are an integral part of the existing gaming platform. For example, earthquakes occur periodically, which generate structural damage and fire. Players must learn to cope with and manage such disasters.



planetary reservoirs, and of the time and space scales of evolution of earth processes were also exposed. Students summarized parametric and sensitivity analysis experiments and researched the literature to scientifically explain what they were observing through *SimEarth* experimentation. The student response was overwhelmingly positive and was reflected in substantially improved written reports.

We have just begun using *SimCity* in the freshman class *Earth Resources and the Environment*, to introduce basic concepts of urban functioning and the interaction between the budgetary process, placement of water, energy and waste treatment facilities, pollution, economic and sectoral growth, including population and resource availability projections. Students make choices between infrastructure elements, and systematically analyze changes in growth, resource demands, and pollution in space and time. Data abstracted from the simulations is analyzed offline to introduce basic statistical and model fitting concepts.

While this fledgling effort is a step in the direction of the proposed development and use of

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OPTIMUS, the *SimCity* game alone is just that, a game, and does not have the transparent analytical (or empirical data driven) modeling engines from which students learn with the rigor desired for a meaningful engineering education. The challenge, then, is to extend these gaming platforms to readily incorporate multiple scales/levels of infrastructure analysis and to explicitly include technical content, including analysis of the results of a simulation (for a life cycle or during episodes of flooding events) and concepts of risk and uncertainty. Ideally, one would use a high level programming language to allow students to easily build modules and attach them to the modeling system. Don Hopkins, (<http://www.donhopkins.com/>) creator of *The Sims*, has enthusiastically agreed to work as a consultant to help us design and implement an open courseware system that can meet the challenge. Don has collaborated extensively with Will Wright, the designer of *SimCity*, and David Levitt, the creator of the visual programming language *Bounce*, who has also plans to assist Don in this project.

#### IV. Objectives and Expected Significance

Our specific project objectives are to:

1. Design a 4 year integrated *undergraduate curriculum* for Civil and Environmental Engineering that offers early exposure (freshman and sophomore) to infrastructure and environmental problems using a systems engineering approach.
2. Develop class content including *case studies* inspired by and using data from real situations for three classes to be offered in the first two years of the curriculum.
3. Develop OPTIMUS, an *Open Platform for Teaching Integrated Modeling and Urban Simulation* that facilitates:
  - a. Presentation of current urban infrastructure and environmental problems and data to students;
  - b. Visualization of multiple physical and socio-economic processes in a 3D landscape;
  - c. Scenario analysis of historical and future conditions under specified patterns of

development on a landscape and variable external factors such as climate and economy;

- d. Integration of numerical and statistical modeling and analysis tools, together with “soft” information and rules for a variety of space-time processes; and
  - e. User interaction with and programming of component models at the level of an infrastructure component or globally.
4. Develop several working prototype scenarios for direct use in the newly re-tooled introductory case-based courses.
  5. Systematically evaluate the curricular change to measure:
    - a. its impacts upon enrollment in the civil and environmental disciplines; and
    - b. its effects upon the learning process
  6. Disseminate project results through seminars, web-site materials, a feature magazine article and journal publications.
  7. Make the developed programming platform widely available to the academic and research community.

While the curriculum developed is directly relevant to Columbia University, it is our hope that the design and implementation experiences will be relevant to the academic community as a whole. Several engineering educators (e.g., Jester, 1989; Parker et al, 1990) frustrated with incremental and ad hoc changes in the curriculum have recommended a comprehensive redesign of the undergraduate curriculum from scratch. The call for increased practical relevance, social context, active and cooperative learning and the use of a systems approach to instruction have often been made (Bordogna, 1998). However, Roesset and Yao (2000) note that given the existing faculty reward structure in most universities, the tendency to emulate rather than innovate on the educational front, and the logistics of change, other than a few, well received pilot efforts, a comprehensive

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rethinking of what should comprise a modern curriculum and how it could be implemented has not occurred. Columbia's commitment comes both from a core of faculty who wish to do this, and from administrative support at all levels.

The second objective entails making a data and information driven approach to engineering a reality from the earliest part of the educational process. It calls for public-private sector partnerships with the University to identify representative problems that can be analyzed at different points in the curriculum with varying degrees of sophistication. Case studies derived from faculty consulting experiences have often been used to great benefit in upper division and graduate classes. Here, we seek to extend the model to infrastructure and environment problems that are likely to have considerably greater scope than the usual consulting project. We have already had some success engaging NY City and NY-NJ state agencies and Con Edison in helping develop some ideas, and participate in developing the case studies in a manner that would help with employee training and community outreach.

The development of the modular programming system OPTIMUS is technically the most challenging but rewarding aspect of the project. This is an idea that has intrigued many practitioners, academics and researchers. In past projects we have worked in (<http://www.tmdl.org/software.html>) the GIS framework and integrated existing engineering models. This is consistent with other attempts (<http://networks.geog.uu.nl/publications.html>). Very few have attempted to put together an open, customizable environment for space-time modeling with the panache of a gaming environment and the necessary scientific sophistication. The collaboration with Hopkins and Levitt, who have extensive experience in developing and implementing new programming languages and constructs and specifically in building Cellular Automata based simulators that capture various aspects of physical and human behavior in an engaging way holds promise for a breakthrough in developing an open system for virtual reality explorations in engineering education.

We recognize that our excitement for the development of the materials above has to be tempered by the reality of demonstrating that students educated using these ideas actually enjoy them to the extent of electing our engineering profession of choice, and more importantly are better prepared through an improved understanding of context and problem decomposition and formulation, but also in terms of measurable performance on instruments such as the *Fundamentals of Engineering* exam. The evidence from other studies is encouraging for change from the existing lecture driven model, and there is an emerging literature on how to conduct such assessments. We expect modest methodological contributions to this literature, but hope for significant impacts upon student learning processes.

## V. Approach

The proposed plan for developing and implementing a new curriculum facilitated by OPTIMUS is presented through an illustration of key factors. The structure of a desirable curriculum including its goals is identified first. A comparative discussion of teaching methods is then used to motivate extensive use of simulation, which leads to a conceptual description of the software technology OPTIMUS. Methods for assessing the impact of the proposed curricular adjustment are then reviewed. Plans for making our work available to the wider community are enumerated, and finally the project management structure is outlined.

### V.1 Curriculum Design

**Engineering Education in the 21<sup>st</sup> Century.** Discussing baccalaureate engineering education Bordogna (1998) comments that it has been too focused on “*reductionistic science*” and too little on holistic elements. He notes that this approach drives many capable students away at the front end. For the 21<sup>st</sup> century, he argues that the engineer's essential role is an integrative one, and education needs to foster the following capabilities:

| 20 <sup>th</sup> Century Emphasis | 21 <sup>st</sup> Century Emphasis |
|-----------------------------------|-----------------------------------|
| <i>In-depth thinking</i>          | <i>Functional Thinking</i>        |
| <i>Abstract Learning</i>          | <i>Experiential Learning</i>      |
| <i>Reductionism</i>               | <i>Integration</i>                |
| <i>Develop Order</i>              | <i>Correlate Chaos</i>            |
| <i>Understand Certainty</i>       | <i>Handle Ambiguity</i>           |
| <i>Analysis</i>                   | <i>Synthesis</i>                  |

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| 20 <sup>th</sup> Century Emphasis | 21 <sup>st</sup> Century Emphasis |
|-----------------------------------|-----------------------------------|
| <i>Research</i>                   | <i>Design/Process</i>             |
| <i>Solve Problems</i>             | <i>Formulate Problems</i>         |
| <i>Develop Ideas</i>              | <i>Implement Ideas</i>            |
| <i>Independence</i>               | <i>Teamwork</i>                   |
| <i>Techno-Scientific Base</i>     | <i>Social Context/Ethics</i>      |
| <i>Engineering Science</i>        | <i>Functional Core of Eng.</i>    |

He argues that 20<sup>th</sup> century education emphasized the left column, and attention is now needed to the items in the right column. Developing contextual understanding, an ability to create, operate and sustain complex systems, skills for lifelong learning, and communication skills constitute the functional core of engineering.

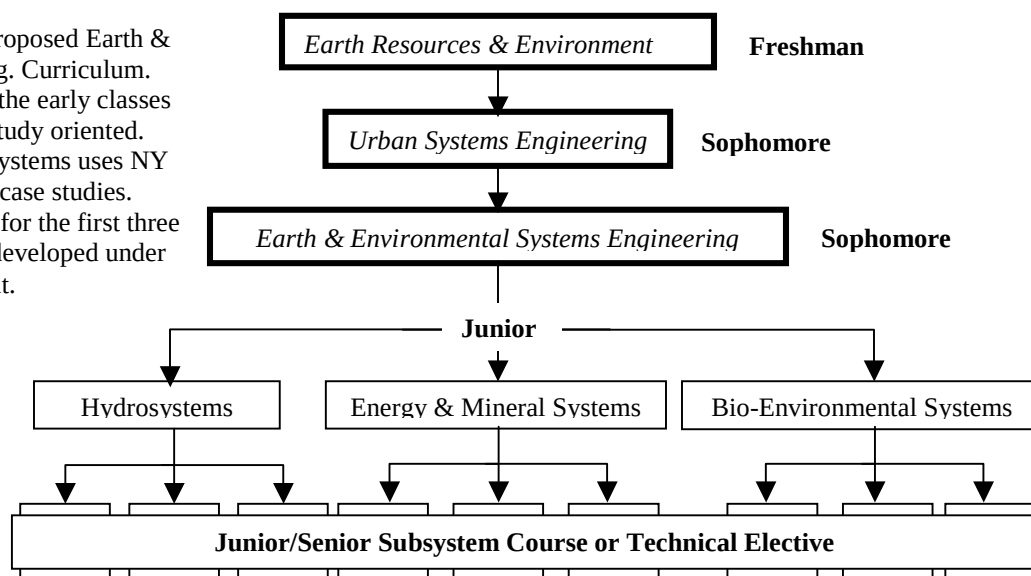
**Engineering Education at Columbia.** The current Columbia curriculum invests the first two years largely in science and humanities courses that provide valuable but seemingly disjoint knowledge. The subsequent years provide exposure to unit process skills culminating in a capstone design project. For example, a student may learn to analyze and design structural elements, and in an advanced class procedures for structural response to acceleration (earthquake), but they will rarely be exposed to the issue of the attendant water, electric, transportation system failures and fires. Yet, with existing infrastructure the synthesis of a response to such a hazard across all these elements is likely to be more valuable.

A keystone of our plan is to focus on the solution of infrastructure and environmental problems with an early exposure to systems analysis and problem formulation concepts using

a case study approach to instruction. Unit process models are introduced in this context, rather than as primary building blocks. This approach necessitates a significant change in the existing curriculum (and in the method of instruction) as illustrated *in concept* for the EEE program in Figure 1. A similar structure will evolve for the CEEM program. The existing curriculum is marked by skill or process based classes. The new curriculum is anchored by a sequence of classes in the first three years that progressively expose the students to a variety of EEE problems in a case study mode. The proposed curriculum gets progressively specialized as one moves to the higher grades, offering a student the opportunity to explore a subsystem given an understanding of the larger context of the problem. For instance, hydrology is now taken in the senior year with no prior context. In the new curriculum, the student will have experienced hydrologic analyses several times by the time a formal hydrology class is taken. This will allow us to teach hydrology in a much more sophisticated manner.

The systems approach is first introduced at the freshman level, as a formalism for identifying the salient aspects of the problem, defining key state and control variables, and identifying the type of scenarios to generate given the objectives or goals of the stakeholders. An exposure to the complexity of the coupled systems, the interplay of scales, and the emergent uncertainties is provided. The role of different bodies of knowledge (e.g., physics, statistics, economics,

**Fig. 1** Proposed Earth & Env. Eng. Curriculum. Each of the early classes is case study oriented. Urban Systems uses NY City for case studies. Content for the first three will be developed under this grant.



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sociology) emerges naturally. Analysis of infrastructure systems for the mega-city comes next. The third class in the sequence reinforces systems engineering skills through the application of numerical, optimization and Monte Carlo simulation tools. Sub-system theme classes follow. The details of the new curriculum will evolve through discussion amongst the PIs, the larger campus community, and participating external agencies.

#### **V.2 Teaching Methodologies**

Educational specialists are critical of passive, lecture based learning methods. We often hear from students that they learn more from peers or from one on one interaction with the professor than they do from lectures. Problem formulation seems to require more of such interactions than the solution of book problems. Consequently, *active and cooperative learning* (ACL) methods are recommended (Johnson et. al., 1991; Mills and Cotell, 1998; <http://coalition.tamu.edu/Zope/acl>, <http://chemistry.calstatela.edu/Chem&BioChem/active/index.htm>; <http://www.udel.edu/cte/pbl.htm>; <http://virtual.clemson.edu/OTEL/resources/cooperative.htm>; <http://www.active-learning-site.com/sum1.htm>)

While education researchers have been concerned with developing and testing the effectiveness of specific ACL methods, our interest is primarily in their *selective* and *effective* application. Over the years we have had direct experience in using such methods and offer the following perspective and direction.

**Complex Systems and Simulation.** Complex systems pervade every nook and cranny of daily life, from the patterns of traffic in urban transport networks to the movement of currents in the ocean. These systems form the core of advanced study in civil and environmental engineering. These processes are fundamentally different from the simple systems that all too often constitute the focus of introductory courses in engineering. Simple systems generally involve a small number of individual elements with relatively weak interactions between them, or they are systems like enclosed gases or distant galaxies, composed of such vast numbers of objects that we can employ statistical averaging techniques to study their behavior. Complex systems, on the other hand, involve a medium-sized number of agents, such as drivers or molecules. Moreover, the

agents in a complex system make their decisions and update their rules for actions on the basis of local, rather than global information. What makes a system a system and not simply a collection of elements is the connection and interactions among the individual components of the system, as well as the effect these linkages have on the behavior of the components.

Fortuitously, just at the moment in history when the unforeseen behavior of these systems is bringing them most forcefully to our attention, technology has presented us with a marvelous tool with which to probe and learn about their mercurial nature. This tool, of course, is the digital computer simulation. Computer simulation can be a laboratory for studying the informational structure of complex systems. The creation of silicon surrogates of real-world complex systems allows us, teachers and students, to perform controlled, repeatable experiments. We can play myriad sorts of what-if games with genuine complex systems.

These computer environments illustrate interactive whole systems, organize and integrate complex skills, and show how individual actions affect complex systems. Such simulations enhance learning through visualization, experimentation, and creativity of play. Increased learning occurs by problem solving in a complex interactive multidisciplinary environment and by “seeing” causal relationships between individual action and whole systems. The broader implications of using such simulations in civil and environmental engineering classrooms are for students to become more effective learners and thinkers enabling them to make connections across the curriculum.

#### **Promise of Simulation in Education.**

Simulation is a cornerstone of discovery learning-styles where students are immersed in an environment that encourages them to infer concepts and to actively engage a problem (Sonwalkar, 2001). More simply, students “learn by doing”, and computer simulations have been used widely in education to provide some experiential learning about the interplay of complex forces in a range of subjects: archaeology, architecture, science, and history. As discussed previously, CCNMTL has

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demonstrated the benefits of simulations in higher education by designing, building, implementing, and evaluating simulations for environmental studies, medical education, and journalism.

We have found simulations to provide a unique environment for exploring new concepts, for gaining an understanding of the interplay between complex phenomena, and for the construction of simplified working models of topics under study. Each of these simulations is used to model some aspect of the world, which users or learners can then explore in a way that is safe and controlled. Parameters are to be manipulated, and the answers to ‘what if?’ questions tested.

Using computer simulations, students are able to gain knowledge in a very active, individual way. The most common way in which educational computer simulations are used, in which concrete experience is built up by active experimentation, and the proposal and modification of hypotheses plays an important role, is exactly the type of activity supported by the existing theories of learning by discovery constructivism. This is the reason why many authors (such as Kolb and Goldman) have emphasized the importance of simulations in teaching and learning.

Simulations are a prime example of constructivist conceptions of learning that claim that in the absence of a real-world, relevant context for learning information, the information is less meaningful, that is, it lacks a referential hierarchy. Too often, learning materials or environments are stripped of contextual relevance. Learners are required to acquire facts and rules that have no direct relevance or meaning to them, because they are not related to anything the learner is interested in or needs to know. We believe that useful knowledge is that which can be transferred to new situations. The most effective learning contexts are those which are problem- or case-based, that immerse the learner in the situation requiring him or her to acquire skills or knowledge in order to solve the problem or manipulate the situation. So, instruction needs to provide contextually based environments that are meaningful to the learners.

#### **Putting Simulation into Practice.**

Technology can facilitate our goal of producing reflective practitioners of engineering that are prepared for the engineering tasks of the 21<sup>st</sup> century. Computers provide a conversational environment in which the learner can apply knowledge to problems and consider their actions as reusable events. Learners can control their learning, learn from others and develop reflection in action and reflection on actions as metacognitive skills.

By placing students in teams and requiring them to investigate real-world problems utilizing a sophisticated simulation engine, we expect to introduce them to Bordogna’s new requirements for engineering by increasing their level of comfort with synthesis and integration activities, experiential learning, teamwork and correlating chaos. In addition, one of the key issues in engineering education is the recruitment and retention of underrepresented minorities and women, we believe that well-constructed simulation programs that are rich in data and that have been created according to the constructivist pedagogic could be used to advance these goals.

#### **V.3 Simulation/Gaming Platform: OPTIMUS**

**Desirable Attributes.** In our context, some desirable attributes of a simulator are:

- *Function:* Facilitate the representation of the multi-faceted evolution of the infrastructure and the environment in a region through the identification of interacting subsystems and their institutional framework.
- *Scenarios:* Ability to assimilate historical data and run scenarios using past conditions with changing parameters (e.g., different zoning or river corridor development plan). Ability to run future scenarios with parametrically specified controls, and different global/local features.
- *Processes/Subsystems:* Ability to isolate or define subsystems (e.g., water supply, transportation) and associated processes, and flexibly add subsystems and define their interactions
- *Temporal Structure:* Integrated event (e.g., for natural hazards) and multi-

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decade (e.g., infrastructure life cycle) simulations.

- **Spatial Structure:** Ability to represent realistic city/landscape layouts to make the relationship between the infrastructure and the environment explicit. Ability to make use of extensive spatial data fields available as part of community/agency GIS platforms.
- **Models:** A consistent, modular framework that allows spatial and process interaction to occur, and enables easy substitution of process models, rules or parameterizations. Accommodates interactive economic, physical, chemical, ecologic and social constructs.
- **Processing:** Make space-time fields accessible at various levels of aggregation for “performance” measure computation and statistical analysis.
- **Interface:** It should look “real”. The user interaction should be intuitive and make all analytical features of the system accessible, with a *LEGO* like ability to add features and change representations. Ability to work with components of the system, zoom into the function of a facility (e.g., water treatment plant), save and manipulate simulations.

**SimCity and Cellular Automata.** A gaming system will be used to develop and introduce the concept of a system and interacting subsystem, as well as basic ideas of mathematical and symbolic modeling and their relationship to data. Concepts of risk and uncertainty will be illustrated through the relationship of events generated stochastically, their expected frequencies and the spatial impact. Vulnerability and exposure to natural and environmental hazards will be made explicit in a spatial context.

The *SimCity* gaming environment is appealing as a building block for many of the features we indicate above. It’s a “god game”, in which the player assumes the role of master controller (mayor) and directs the simulation through the design of a city (power, water, waste, education, fire, etc subsystems) and its operation (budgeting, ordinances, environmental investments). Feedback on the mayor’s performance is provided continuously through

spatial maps of pollution, growth measures for residential, commercial and industrial sectors, fires that break out in areas outside the range of designed service, and pop up windows that represent citizen concerns. The player functions as an implicit optimizer seeking to grow the city to a high degree of affluence and population, accomplished by balancing growth amongst sectors and maintaining a good environment.

Students often have considerable experience with such games and adapt very quickly to interacting with them. *SimCity* lacks the open architecture to add modules and extract state variables.

However, Hopkins (<http://www.catalog.com/hopkins/simcity/x11-demo.html>) has demonstrated how to make this game extensible to a multi-player, multi-platform setting, and to make the state variables accessible. The multiple roles allow student teams to come in as a city public works engineer, a water systems manager etc., and explore what is involved in designing the system and interaction with the institutional and regulatory setting. A limited capability to import GIS terrain and land use data directly into SIMCITY already exists. Palettes to change the landscape, easily add infrastructure elements and the like already exist. They could be enhanced to enable realistic specification of components (e.g., conveyance pipes).

The various *SIMgames* use Cellular Automata (CA) as the underlying modeling structure, which allows spatial interaction and space-time evolution. Classical models for fluid and heat transfer based on partial differential equations are readily recast as CA models, enabling the instructor to start introducing these concepts early in the curriculum. CA models are also used extensively for spatially extended ecological, environmental and social modeling. Local (in space and time) rules are used to identify the evolution of the system from one time to the next. The CA framework appears to be very useful as a building block for spatial models and for teaching material on interacting subsystems. Specifically, the ability to model processes at multiple scales is already illustrated by *SimEarth* (focuses on the planet), *SimCity* (focuses on the city) and *SimTower* (focuses on a building). While these games are not linked, they

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provide a proof of concept for developing an environment in which pseudo-realistic, multiple scale modeling of such systems can be done.

High-level visual, programming languages such as *VPython* provide robust, visual programming capabilities very efficiently (see Box 2). Foremost, these languages permit production of graphic, visual sequences with surprising ease. Additionally, their simplicity makes it possible to involve students in the development of visualization modules to enhance the simulator rather than simply having the students utilize it. Hopkins, Wright and Levitt have demonstrated the power of such languages in simulation games already mentioned. Hopkins and Levitt's participation in this project supplies

the requisite technical and intellectual knowledge necessary to develop the simulator envisioned.

**Integration.** Integrating a visual platform with data and models that represent the socioeconomic, natural and physical fabric of an urban center holds substantial promise for engineering instruction. Students can be exposed to various scenarios within a realistic setting and then be given the opportunity to investigate, design and implement changes to the natural and built environment; subsequently, they will also have the chance to observe the effects of the strategies that they implement. *SimCity*'s spatial and visual platform provides a ready vehicle for integrating data sources and models to represent the social and physical systems that embody an urban center. Once links are established between three-dimensional images of systems or components and databases or models that store information or represent physical behavior, the possibilities as an instructional tool are quite exciting.

Interaction with the tool could take a number of forms. Students could access a database from the visual simulation window by selecting an object in the 3D environment of an urban neighborhood. This could highlight 2D GIS maps and data tables in a separate window. Students could then query and analyze demographic, economic or technical data in the region of interest. In addition, students could investigate and study changes in the urban landscape by "picking" objects from a 3D scene. Once selected, an object could be removed from the scene (simulating, for example, the removal of a building from a lot) or alternative models could be substituted for the object. Students could easily design and review different options for a particular site or region.

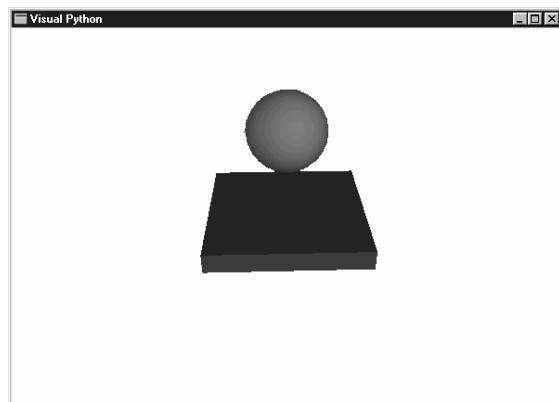
**Example Scenarios.** Most newcomers to engineering, and more broadly the general public, are interested in failures of the built environment rather than successes (which are often taken for granted). Two of the proposed simulation scenarios include urban earthquake and flooding disasters, which possess characteristics that align with the project's pedagogical objectives. Each scenario demonstrates the *interdependency* of infrastructure systems, the effects of *spatial*

#### **Box 2 – High-Level Programming Example**

Below is an entire *VPython* program that produces 3D animation of a red ball bouncing on a blue floor. Coding this effect in languages such as C++ would require substantially more code.

##### **from visual import**

```
floor = box (pos=(0,0,0), length=4, height=0.5,
            width=4, color=color.blue)
ball = sphere (pos=(0,4,0), radius=1,
              color=color.red)
ball.velocity = vector(0,-1,0)
dt = 0.01
while 1:
    rate (100)
    ball.pos = ball.pos + ball.velocity*dt
    if ball.y < ball.radius:
        ball.velocity.y = -ball.velocity.y
    else:
        ball.velocity.y = ball.velocity.y - 9.8*dt
```



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distribution, and the *uncertainty* of natural hazards. For instance, during an urban earthquake disaster, structures will suffer varying degrees of damage or potentially total collapse depending on the structure type, location relative to the epicenter, the soil conditions at the site of the structure and many other factors. An existing software tool HAZUS, which was developed by FEMA as their “Natural Hazard Loss Estimation Methodology” for the United States, would be linked to the basic SIM package with which the students interact. The figure below illustrates just one example of the type of damage distribution output available from HAZUS for a scenario earthquake with an epicenter located several miles from New York City. (Here the darker colors represent higher concentrations of direct dollar losses). HAZUS can also generate loss of life estimates.



The basic data which went into this simulation, performed by our colleague Prof. George Deodatis, were the soil types, distribution of the building stock (e.g., single story, high-rise, mid-rise apts. etc.). Such data would remain fixed, and students could study the consequences of different magnitudes and earthquake epicentral locations relative to the city being considered. More importantly, basic engineering concepts can be introduced: such as what pre-

emptive measures designers could take to mitigate the effects of such a natural disaster.

Modifiable design parameters would include:

- Increase Structural Ductility
- Increase Structural Stiffness
- Pursue Base Isolation Strategies
- Relocate and/or Condemn Inadequate Systems - this is a site-specific issue.
- Add Passive or Active Dynamic Control Systems

Naturally costs would be associated with these design choices, and students could evaluate the cost/benefit of the design choices with the expected dollar losses estimated by the HAZUS loss estimation kernel. This result emphasizes the insurance and public policy aspects of basic design issues in civil engineering.

Urban flooding disasters share some of the characteristics of an earthquake, but event effects and mitigation strategies are different. Whereas the epicenter of an earthquake is uncertain, the boundaries of a flood event are not. This bounds the engineering problem to a degree, so students would have the opportunity to select a discrete set of mitigation strategies, normally control and/or protection measures, and then observe the impacts of these strategies upon resulting flood damage at varying levels of flood magnitude and implemented cost. Mitigation strategies, however, often have unintended consequences, so the simulation platform is a ready vehicle to illustrate the impacts of designed solutions to students. This will enrich their appreciation of the significance of sound engineering analysis coupled with situational awareness and sensitivity. Regions that are flood-prone such as Sacramento, CA are already included within SimCity (shown earlier in Box 1). Such a region would be a natural starting point for development of the type of scenarios described.

#### **V.4 Assessment and Evaluation**

Once we have implemented the prototype simulation in the classroom, we must insure that the effects intended have, in fact, been achieved. Obvious metrics that are easily observed such as student enrollment, retention and graduation rates will be tracked after implementing the adjustments proposed, but assessing the impacts of OPTIMUS and the curricular change upon student learning processes is far more important,

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as well as more difficult. By design, however, we shall introduce the two sophomore level classes, Urban Systems Engineering and Earth & Environmental Systems Engineering into the curriculum in the spring of 2003 before the OPTIMUS prototype is developed. This permits the assessment of student performance and learning **without** the simulator and **with** the simulator to compare results. Students will confront similar case-based problems in each version of the course, but students in the first version will not have the opportunity to visualize scenarios using OPTIMUS. We intend to deploy the assessment tools subsequently described to evaluate student performance.

**Assessment Tools.** Final tests commonly determine the effect of the teaching. However, alternatives such as group presentations and written reporting are preferable because they reinforce the learning process. Student groups can present their results and conclusions before the class which permits discussion of any differences. A great advantage of group presentation is that during this phase the teacher can control the result of the learning process and make adjustments (e.g., by correcting any wrongly drawn conclusions). Written reporting is a second alternative that requires students to more formally articulate results.

Group presentation and written reporting force the student to set out their observations in an organized format, and the organization of learning material can (again) reinforce the learning process. In addition, group presentation and written reporting are both important skills to include in engineering courses.

Apart from the tools mentioned above, a second type of evaluation is possible where the aim is to gauge the quality of the education given and, if necessary, to improve it. Evidence is collected from three sources: the learning effect, controlled using the methods mentioned above; the observations of the teacher and teaching assistants (if present); and a users' analysis (student opinions of the teaching provided). The simplest way of performing a users' analysis is to provide a questionnaire to all the students and follow this up with a focused task analysis with selected users. Using these techniques, information can be acquired about the clarity of

materials, the relation of materials to other elements of instruction, the degree of difficulty relative to the target group, and the desirability of chosen teaching methods.

#### **V.5 Plans for Knowledge Dissemination**

Making the working prototype accessible and reporting results to the broader academic and research community are significant dimensions of the project. A project web-site will serve as the primary portal of communication. Activities, progress and papers will be posted to the site regularly. In addition, members of the academic community will be invited to half-day seminars that will be scheduled periodically. Formal discussion of the project, its results and lessons learned shall be presented at appropriate conferences and published in relevant journals. Publication of a feature article in *PRISM* magazine of the American Society of Engineering Education (ASEE) and presentation of project results at ASEE's annual conference will be high priorities.

OPTIMUS also has a clear use within secondary education programs. Students in their final years of secondary education could utilize the tool to obtain exposure to the context of civil and environmental engineering problems and the roles that engineers play in solving them. Such a computer environment is quite familiar to most high school students, so this is a natural vehicle for stimulating their curiosity of the engineered systems and physical behavior. Outreach activities with high schools in neighboring communities within Harlem and the Bronx are planned. In addition, we will make OPTIMUS available to the Salvadori Center, a non-profit organization based in New York City dedicated to using the built environment as a vehicle for educating K-12 students in topics such as physics, engineering and history.

#### **V.6 Project Management Structure and Institutional Support**

**Key Project Participants.** The project is a collaboration of personnel from three key groups: (1) CEEM and EEE faculty, (2) CCNMTL professional staff, and (3) expert programming consultants. CEEM & EEE faculty members Upmanu Lall, Michael Garvin and Andrew Smyth shall act as the project's principal investigators. They shall guide curriculum

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revision and serve as content experts for the development of the simulation/gaming platform. Lall is a Professor in EEE, and he is currently leading a department-wide effort to reconfigure undergraduate courses. Garvin and Smyth are assistant professors in CEEM with expertise in horizontal and vertical infrastructure systems, and both were recently hired by the department. Currently, Garvin is responsible for instruction of the department's undergraduates in the fundamentals of civil infrastructure management while Smyth is responsible for instruction of the department's undergraduates in the mechanics of solids and structural design.

Peter Sommer is the Director of Education for Projects and Services of CCNMTL. He will act as the project's pedagogical expert and will direct the development of the simulation/gaming platform. He and his staff have successfully deployed simulation tools in both undergraduate and graduate courses at Columbia University. Additional staff of CCNMTL who have developed simulation tools such as *Brownfield Action* and *Heart Simulator* are also part of the project team. Don Hopkins has substantial experience in the design and production of urban simulation platforms. As a project consultant, he will provide expertise and guidance in design concepts, programming techniques and gaming strategies in urban simulation.

**Advisory Committee.** An advisory committee of faculty members from throughout Columbia University will oversee the project's evolution; committee members will provide technical and content input as needed. Regular updates to the committee will disseminate project

results throughout the University thus providing further opportunities to integrate active learning and simulation tools into other curricula. The committee will include representatives from EEE, CEEM, and other departments throughout the School of Engineering & Applied Science, as well as the Departments of Chemistry, Earth & Evolutionary Biology, Economics, Physics, Psychology, and Urban Planning.

**Project Execution.** Critical activities and key milestones are depicted in the Gantt Chart below. During initial stages, the project team will further define simulation scenarios and evaluate data sources and models for inclusion in OPTIMUS. Prior to initiating full-scale development activities, the consulting team led by Hopkins will conduct a programming orientation to provide an overview of existing features in *SimCity* and higher level programming methodologies in urban gaming/simulation. Once platform development begins, this consulting team will remain on-call to assist as necessary. Development activities will then begin in earnest to prepare a prototype by late fall of 2003.

The Urban Systems Engineering and Earth Resources Engineering courses will be introduced into the curriculum in the spring of 2003 in the absence of the simulator. In late spring of 2004, members of the academic and research communities within Columbia University and elsewhere will be invited to a half-day seminar to demonstrate the simulator and to discuss project results. Finally, formal discussion of project activities and results will be published.

| Activity                               | 2002         | 2003                                                         |              | 2004         |
|----------------------------------------|--------------|--------------------------------------------------------------|--------------|--------------|
|                                        | Fall         | Spring                                                       | Fall         | Spring       |
| Define Scenarios                       | XXXXXXXXXXXX |                                                              |              |              |
| Identify & Evaluate Data and Models    | XXXXXXXXXXXX |                                                              |              |              |
| Assemble Programming Team              | XXX          |                                                              |              |              |
| Programming Orientation                | XXXXXXXXXX   |                                                              |              |              |
| OPTIMUS Development                    |              | XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX |              |              |
| Introduce USE & ERE Courses            |              | XX                                                           |              |              |
| Prototype Alpha Version Prepared       |              |                                                              | XX           |              |
| Test Alpha Version                     |              |                                                              | XXXXXXXXXXXX |              |
| Utilize Prototype in USE & ERE Courses |              |                                                              |              | XXXXXXXXXXXX |
| Dissemination Seminar                  |              |                                                              |              | XX           |
| Report Results                         |              |                                                              |              | XXXXXXX      |

Notes:

UIS = Urban Systems Engineering

ERE = Earth Resources Engineering